

Book 4: Hatchery Management & Breeder Operations

A Comprehensive Technical Manual for Commercial Egg Incubation & Parent Stock Management

PoultryWise Operational Manuals • Standard Operating Procedure (SOP) Reference

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CHAPTER 1: PARENT STOCK & BREEDER MANAGEMENT

The foundation of high hatchery performance begins at the breeder farm level. Producing top-tier day-old chicks (DOCs) requires maintaining healthy, highly fertile parent stock through strict nutrition, lighting, and physical body weight management.

- **Male-to-Female Ratio:** Maintain 1 male per 9 to 10 females (10% ratio) in broiler breeders, adjusting as males age to preserve mating efficiency.
- **Body Weight Uniformity:** Weekly weighing is required to keep flock uniformity above 85%. Uniform pullets mature simultaneously, leading to consistent egg size and hatching windows.
- **Breeder Nutrition:** Feed diets rich in essential vitamins (E, A, and biotin) and micro-minerals (Zinc, Selenium) to support optimum embryo viability and shell structural integrity.

CHAPTER 2: HATCHING EGG COLLECTION & SELECTION

Not all eggs laid on a breeder farm are suitable for incubation. Proper selection directly impacts hatchability percentage and chick vigor.

Selection Criteria	Optimal Standard	Reject / Discard Condition
Egg Weight	52g – 68g (depending on flock age)	< 50g (too small) or > 72g (double-yolk potential)
Shell Quality	Smooth, clean, uniform calcium structure	Cracked, micro-fissured, thin-shelled, or misshapen
Cleanliness	Nest-laid clean eggs	Floor eggs, heavily manure-soiled, or wet eggs
Collection Frequency	4 to 6 times daily minimum	Infrequent collection leading to nest overheating

CHAPTER 3: STORAGE, HANDLING & PRE-INCUBATION

After collection, hatching eggs must be cooled rapidly to halt embryonic development at the blastoderm stage until they are loaded into setters.

- **Physiological Zero:** Eggs must be stored below 20°C (68°F) to suspend cellular division.
- **Short Storage (< 7 Days):** Maintain temperature at 16°C - 18°C with 75% - 80% relative humidity (RH).
- **Extended Storage (> 7 Days):** Lower storage temperature to 12°C - 14°C . Implement SPIDES (Short Periods of Incubation During Egg Storage) to preserve hatchability.
- **Pre-Warming:** Warm eggs gradually at 22°C - 24°C for 12 hours prior to setting to prevent condensation ("egg sweating") and thermal shock.

CHAPTER 4: INCUBATION PARAMETERS & SETTER OPERATIONS

Modern commercial hatcheries rely on single-stage or multi-stage incubation systems to deliver precise environmental conditions during the first 18 days of incubation.

Critical Setter Operating Parameters (Days 1 to 18)

- **Temperature:** Maintain strictly at 37.5°C - 37.8°C (99.5°F - 99.8°F). Eggshell temperature (EST) should stay at 37.8°C (100°F).
- **Relative Humidity:** 50% - 55% RH. The target is an egg weight loss of 11% - 12% by Day 18 due to moisture evaporation.
- **Turning Angle & Frequency:** Turn eggs 45 degrees relative to the vertical axis every 60 minutes to prevent embryo adherence to the shell membrane.
- **Ventilation / CO₂:** Keep CO_2 below 4,000 ppm. Proper airflow provides oxygen and removes metabolic heat as embryos develop.

CHAPTER 5: EMBRYONIC DEVELOPMENT & CANDLING

Monitoring embryonic progression allows early detection of fertility issues and machine calibration problems.

- **Candling Timeline:** Performed between Day 10 and Day 18 (typically during transfer to hatchers).

- **Clear Eggs Identification:** Remove infertile eggs and early embryonic deaths ("clears") to prevent bacterial contamination and heat loss in hatchers.
- **Transfer Phase (Day 18):** Move viable eggs from setter trays into hatcher baskets carefully to minimize mechanical damage to embryonic blood vessels.

CHAPTER 6: HATCHING & PULLING CHICKS

During the final 3 days (Days 18 to 21), incubation parameters shift focus to support lung ventilation, internal pipping, and shell emergence.

Phase	Target Temperature	Target Humidity (RH)	Key Objective
Transfer (Day 18)	$36.8^{\circ}\text{C} - 37.0^{\circ}\text{C}$	55%	Embryo repositions for pipping
Pipping Peak (Day 20)	36.5°C	65% - 70%	Moisture prevents membrane drying
Drying Phase (Day 21)	36.0°C	50%	Chicks dry, down becomes fluffy

CHAPTER 7: CHICK PROCESSING, SEXING & GRADING

Once hatch is complete (typically 504 hours from setting), chicks are processed prior to farm dispatch.

- **Quality Grading (Pasgar Score):** Evaluate chick length, navel condition, legs, reflex response, and yolk sack absorption. High-quality chicks are active, clean, and free of defects.
- **Sexing Techniques:** Feather sexing (fast/slow-feathering lines) or vent sexing performed by certified hatchery specialists.
- **Vaccination & Beak Trimming:** Spray vaccination for Infectious Bronchitis (IB) and Newcastle Disease (ND); automated subcutaneous or in-ovo vaccination for Marek's disease. Infrared beak treatment (IBT) when specified for layers.

CHAPTER 8: HATCHERY BIOSECURITY & SANITATION

Hatcheries provide ideal conditions for bacterial and fungal growth if strict hygiene standards are not enforced.

- **Unidirectional Air & People Flow:** Enforce strict movement controls from "clean" areas (egg receiving, setters) to "dirty" areas (chick processing, washer rooms).
- **Disinfection & Fogging:** Continuous automated fogging using formaldehydes or specialized quaternary ammonium/peracetic acid compounds in setter/hatcher halls.
- **Waste Handling:** Immediate disposal and maceration of unhatched shell fragments, infertile eggs, and culled chicks following animal welfare protocols.

CHAPTER 9: QUALITY CONTROL & TROUBLESHOOTING

Systematic residue analysis on unhatched eggs helps identify key drivers of low hatchability.

Residue Analysis Diagnostic Guide

- **High Early Dead (< 4 Days):** Often indicates improper egg storage, severe pre-warming thermal shock, or poor flock nutrition/fertility.
- **High Mid Dead (Days 5–14):** Suggests nutritional deficiencies (e.g., Vitamin B12, Zinc) or contamination issues.
- **High Late Dead (> 15 Days):** Points to overheating, inadequate ventilation, improper moisture loss, or incorrect turning angles.
- **Unabsorbed Yolk Sack / Wet Navels:** Caused by low setter temperatures or excessively high humidity during late incubation.

CHAPTER 10: ECONOMIC METRICS & HATCHERY KPIS

To ensure commercial sustainability, hatchery performance must be continuously tracked against industry benchmark key performance indicators (KPIs).

Performance Metric	Calculation Standard	Industry Target KPI
Fertility Rate (%)	$\left(\frac{\text{Fertile Eggs}}{\text{Total Eggs Set}} \right) \times 100$	$\geq 95\%$
Hatchability of Total (%)	$\left(\frac{\text{Total Chicks Hatched}}{\text{Total Eggs Set}} \right) \times 100$	$\geq 85\%$
Hatchability of Fertile (%)	$\left(\frac{\text{Total Chicks Hatched}}{\text{Fertile Eggs}} \right) \times 100$	$\geq 90\%$
First-Grade Chick Yield	$\left(\frac{\text{Saleable Grade 1 Chicks}}{\text{Total Hatched}} \right) \times 100$	$\geq 98\%$

End of Book 4 — PoultryWise Knowledge Series • Hatchery Management & Breeder Operations